

Run 02 | OME

Global (including India)

IV only

Target Opportunity-motivated entrepreneurship; binary 1 vs. 0
Sample 1,342,884 respondents **Coverage** 89 countries; 2015–2022
Positive cases 100,087 (7.45%) **Case time** 37m 21s
Train / val. / test 913,163 / 161,145 / 268,576 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.
Fraser EFW predictors (7): Overall economic freedom, Economic freedom quartile, Size of government, Legal system and property rights, Sound money, Freedom to trade internationally, Regulation. Total independent variables: 20.
Controls (0): None. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: Random Forest (0.9759). Highest F1: Random Forest (0.6960). The all-negative accuracy baseline is 92.55% (full sample).

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
Random Forest	0.9759	0.7124	94.44	58.71	85.43	0.6960	2m 08s
XGBoost	0.9758	0.7048	94.72	61.00	80.95	0.6957	22s
CatBoost	0.9751	0.6961	94.37	58.37	85.24	0.6930	1m 49s
H2O GBM	0.9735	0.6744	94.24	57.84	83.99	0.6851	48s
Lasso	0.9714	0.6424	94.04	56.89	82.64	0.6739	9m 16s
Elastic Net	0.9714	0.6424	94.04	56.90	82.63	0.6739	10m 58s
Logistic Regression	0.9714	0.6421	94.05	56.95	82.68	0.6744	1m 12s
Ridge	0.9711	0.6391	94.04	56.91	82.73	0.6743	7m 14s
Decision Tree	0.9681	0.5998	93.81	56.18	76.74	0.6487	10s
Naive Bayes	0.9079	0.3667	88.80	37.36	74.30	0.4972	2m 59s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	34,148.88	44.92
2	Established business owner	12,670.24	16.67
3	Size of government	2,857.27	3.76
4	Legal system and property rights	2,416.75	3.18
5	Freedom to trade internationally	2,168.64	2.85
6	Self-efficacy	2,166.51	2.85
7	Sound money	2,090.03	2.75
8	Overall economic freedom	2,087.71	2.75

Share is the percentage of total forest impurity importance across all 38 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Fraser EFW predictors within the same forest

Rank	Fraser EFW predictor	Raw importance	Share%
1	Size of government	2,857.27	3.76
2	Legal system and property rights	2,416.75	3.18
3	Freedom to trade internationally	2,168.64	2.85

Fraser EFW subset ranked by raw importance; shares use total forest importance, not the Fraser EFW subtotal.

Source: Completed Objective 3 batch, 29 September 2026; 02_obj3_global_iv_OME. Scores and importance are taken directly from saved CSV outputs.